

3. + MPI / beam remnants

standalone MPI: event_MPI.lhe

211 particles drawn (marker size $\sim \log p_T$; label = p_T [GeV] for $p_T > 5$ GeV), 17 with $|\eta| > 6$ drawn on the border (x)

b quark (status 62): $p_T = 78.29$ GeV, $\eta = -1.38$, $\phi = -2.73$

$\bar{b}$ quark (status 62): $p_T = 72.50$ GeV, $\eta = -3.42$, $\phi = 0.57$

net $b + \bar{b}$: $|\vec{p}_T(b) + \vec{p}_T(\bar{b})| = 13.44$ GeV ($p_x = -10.82$, $p_y = 7.96$), $\Sigma|p_T| = 150.79$ GeV, $\Delta\phi(b, \bar{b}) = 2.98$

ISR (34 partons): $|\Sigma\vec{p}_T| = 41.39$ GeV, $\Sigma|p_T| = 100.71$ GeV

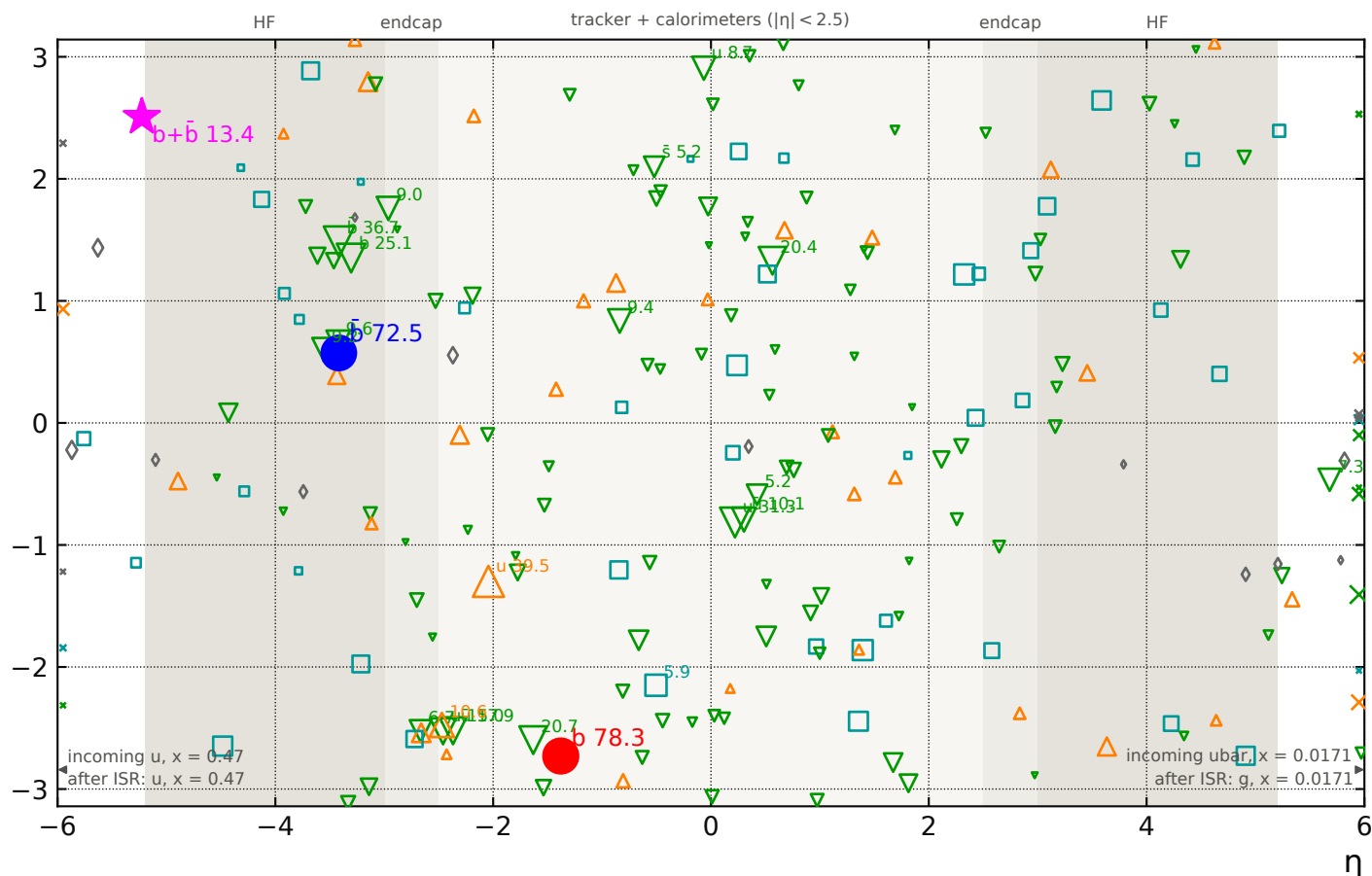
FSR (114 partons): $|\Sigma\vec{p}_T| = 38.33$ GeV, $\Sigma|p_T| = 358.40$ GeV

MPI (45 partons): $|\Sigma\vec{p}_T| = 9.05$ GeV, $\Sigma|p_T| = 78.95$ GeV

beam remnant (16 partons): $|\Sigma\vec{p}_T| = 5.88$ GeV, $\Sigma|p_T| = 8.95$ GeV

all 211 final-state partons: $|\Sigma\vec{p}_T| = 0.01$ GeV (the two protons carry no p_T , so the sum must vanish)

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particle origin (number drawn)

- b quark (1)
- $\bar{b}$ quark (1)
- ★ $b + \bar{b}$ (vector sum)
- △ ISR (34)
- ▽ FSR (114)
- MPI (45)
- ◇ beam remnant (16)

45 secondary scatterings

marker size:

- $p_T = 1$ GeV
- $p_T = 10$ GeV
- $p_T = 100$ GeV

stages:

1 hard process

2 ISR + FSR

3 MPI / remnants

4 hadronisation

5 hadron decays